

Software Requirements Specification (SRS)

NeedNow Map — Intelligent Crisis Command & Triage Platform

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Document Control & Governance

Revision History

Version	Date	Status	Description	Primary Basis
0.1.0	2026-04-10	Draft	Initial civic incident intake prototype and mock map integration.	Initial hackathon mock
0.2.0	2026-05-15	Alpha	Added Nominatim geocoding and basic Gemini prompt triage.	Early AI integration
0.3.0	2026-06-20	Beta	Integrated OpenStreetMap Overpass live queries and basic Firestore sync.	Geospatial MVP
0.4.0	2026-07-28	Phase 2	Added Canonical Capability Registry, zero-hallucination contact extraction, and SSE streaming.	Phase 2 Architecture
1.0.0	2026-08-29	Final	Full reverse-engineered, codebase-grounded technical specification across all 45+ source modules.	Active Codebase Audit

Epistemic Categorization Standards

Every statement and requirement in this specification is categorized using strict epistemic standards:

- VERIFIED IN CODE** : Directly observed in active source code files (`src/app/**` , `src/lib/**` , `src/components/**` , `src/types/**` , etc.).
- VERIFIED IN CONFIG** : Directly observed in repository configuration files (`package.json` , `tsconfig.json` , `postcss.config.mjs` , `.env.example` , etc.).
- INFERRED** : Reasonably deduced from runtime architecture, interface definitions, or standard library behaviors.
- NOT IMPLEMENTED** : Formally documented as an architectural non-goal, future expansion point, or omitted capability.

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1. Executive Summary & Product Vision

1.1 Executive Summary VERIFIED IN CODE

NeedNow Map is an intelligent crisis command, emergency reporting, real-time investigation, geospatial resource discovery, and emergency dispatch decision-support platform. It bridges unstructured civilian crisis reports with real-world civic and emergency response infrastructure through a deterministic, evidence-grounded AI investigation pipeline.

```

Citizen Incident Intake
↓
Telemetry Evaluation & Location Quality Parsing
↓
Incident-Aware Capability Planning (Positive & Negative Scoping)
↓
Controlled Geospatial Infrastructure Queries (OpenStreetMap / Overpass)
↓
Deterministic Resource Ranking & Zero-Hallucination Contact Extraction
↓
Gemini 2.5 Flash Evidence Reasoning (Structured Schema Validation)
↓
Deterministic Multi-Dimensional Confidence Calculation
↓
Real-Time SSE Telemetry Streaming to Citizen / Operator UI
↓
Incident Persistence & Realtime Cloud Firestore Synchronization
↓
Crisis Command Center Dashboard & Tactical Leaflet Map
↓
Human Dispatcher Override & Response Team Deployment

```

1.2 Decision-Support Paradigm VERIFIED IN CODE

NeedNow Map is **strictly a decision-support system**. The platform's AI engine investigates, correlates real geospatial assets, and recommends operational response units. **Human dispatchers retain 100% exclusive command authority** to acknowledge, override priorities, reassign tactical teams, and resolve incidents. The platform never operates as an autonomous emergency dispatch authority.

2. Problem Statement & Civic Crisis Operational Context

2.1 Core Problem Statement VERIFIED IN CODE

During civic emergencies and disaster events, the critical bottleneck is not a lack of emergency response resources—it is **information chaos, spatial disconnection**, and **delayed decision-making**. Emergency command centers and 911/112 dispatchers are overwhelmed by fragmented, unstructured civilian reports, while traditional Computer-Aided Dispatch (CAD) systems lack automated, evidence-grounded spatial intelligence. Consequently, dispatchers face severe cognitive overload, leading to misclassified priorities, delayed response unit deployment, and tragic loss of life during the critical "Golden Hour".



2.2 Key Operational Pain Points VERIFIED IN CODE

- 1. Unstructured & Fragmented Civilian Intake:** Civic reporting channels receive conversational, panic-driven text (e.g., *"Transformer exploded on 3rd floor, thick black smoke, 6 workers trapped, live wires sparking on the road"*). Dispatchers must manually parse narrative text, estimate casualties, extract locations, and determine required multi-agency capabilities under intense cognitive pressure.
- 2. The "Spatial & Infrastructure Disconnection":** Knowing an incident's coordinates is insufficient; dispatchers need to immediately identify **what real-world infrastructure exists in that operational radius** (nearest high-voltage substations for power isolation, nearest Level-1 trauma centers, fire hydrants, municipal water valve depots). Responders waste precious minutes looking up facility phone numbers or calling facilities that lack required specialized capabilities.
- 3. The "Black-Box AI" & Hallucination Hazard:** Generic AI models (such as unconstrained LLMs) hallucinate non-existent emergency contact numbers, fabricate ambulance dispatch readiness, or arbitrarily declare "95% confidence" with no mathematical backing. Emergency services cannot risk life-safety operations on opaque, ungrounded AI predictions.
- 4. False-Positive Scoping & Misallocated Resources:** Keyword-based triage systems often misroute incidents—such as dispatching swift-water rescue boats to a municipal underground pipe leak, or querying critical ICU generator backups during a standard residential power outage. High-value rescue assets are tied up in non-critical situations, leaving genuine catastrophic emergencies unprotected.
- 5. Lack of Real-Time Investigation Transparency:** When citizens submit reports, they enter a "black box" queue with zero feedback, triggering multiple duplicate panic calls that flood emergency switchboards and congest dispatch queues.

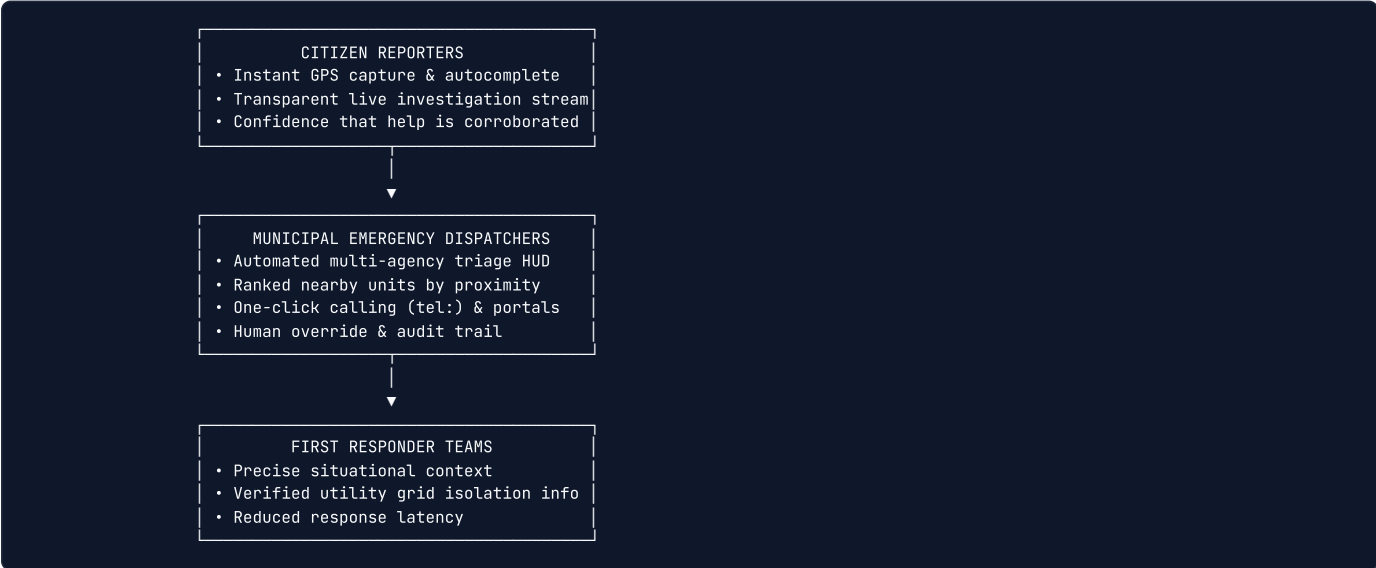
2.3 The NeedNow Map Solution Architecture

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Conventional Emergency Intake	The NeedNow Map Solution
Manual Narrative Parsing: Dispatchers spend 2–5 minutes interpreting unstructured text.	Instant Multi-Stage AI Understanding: Gemini 2.5 Flash extracts structured facts, casualties, and required capabilities in milliseconds.
Siloed Agency Lookups: Dispatchers manually look up fire stations, hospitals, or utility contacts across separate directories.	Live OpenStreetMap Geospatial Discovery: Automatically queries real-world facilities within bounded tactical radii (4–15 km) via Overpass API mirrors.
AI Hallucinations: Generic LLMs invent phone numbers or response times.	Zero-Hallucination Contact Provenance: All telephone numbers, portals, and addresses are extracted strictly from mapped OSM tags with field-level provenance (<code>OSM:phone</code>).
Opaque Confidence Scores: Arbitrary self-scored AI ratings.	Deterministic Multi-Dimensional Confidence Engine: Pure mathematical calculation across Location (<i>L</i>), Narrative (<i>C</i>), Severity (<i>S</i>), and Evidence (<i>E</i>) with auditable point contributors and discrepancy penalties.
Rigid Keyword Matching: Pipe leaks trigger flood rescue teams.	Incident-Aware Positive & Negative Scoping Engine: Strict scoping logic prevents false-positive asset queries (e.g., no boats for municipal pipe bursts unless drowning is explicitly verified).
Black-Box Reporting: Citizen has no insight into response progress.	Real-Time SSE Telemetry Streaming: Live investigation visualizer displays step-by-step geospatial scanning, discovered facilities, and diagnostic reasoning in real-time.
Autonomous Dispatch Risk: Unsafe autonomous AI dispatch.	Decision-Support & Human Command Preservation: AI investigates and recommends; human dispatchers retain 100% exclusive authority to approve, reassign, or override.

2.4 Target Beneficiaries & Operational Impact

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2.5 Executive Value Proposition

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In an emergency, every second wasted searching for the right response team costs lives. **NeedNow Map** is an intelligent crisis command and decision-support platform that transforms chaotic civilian reports into evidence-grounded tactical response directives. By combining **Google Gemini AI** for structured incident understanding, **live OpenStreetMap geospatial queries** for zero-hallucination facility discovery, and a **deterministic mathematical confidence engine**, NeedNow Map provides emergency dispatchers with verified nearby resources, direct calling actions, and tactical recommendations—ensuring human authorities make informed, life-saving dispatch decisions in seconds.

3. Scope, Goals & Non-Goals

3.1 Scope

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- **Citizen Intake:** Web-based interface for reporting incidents with automatic GPS geolocation, OpenStreetMap Nominatim address autocomplete, category tagging, and narrative description intake.
- **Geospatial Capability Discovery:** Automatic spatial search of real emergency facilities (fire stations, hospitals, ambulance depots, police stations, substations, water works) via Overpass API mirrors within bounded tactical radii (4–15 km).
- **Contact Extraction:** Live parsing of OpenStreetMap contact tags (`phone` , `contact:phone` , `website` , `addr:full` , `opening_hours` , `operator`) with field-level provenance (`OSM:phone`).
- **Deterministic Confidence Engine:** Pure mathematical computation of multi-dimensional confidence scores without relying on LLM self-scoring hallucinations.
- **Crisis Command Center:** Restricted operations dashboard featuring split-view geospatial tracking, situational KPI telemetry, 24-hour timeline history replay, and 5-section incident command consoles with direct calling (`tel:`) and portal launch actions.

- **Realtime State Synchronization:** Multi-client synchronization via Cloud Firestore with reference-counted singleton snapshot listeners and fallback circular in-memory store.

3.2 Goals VERIFIED IN CODE

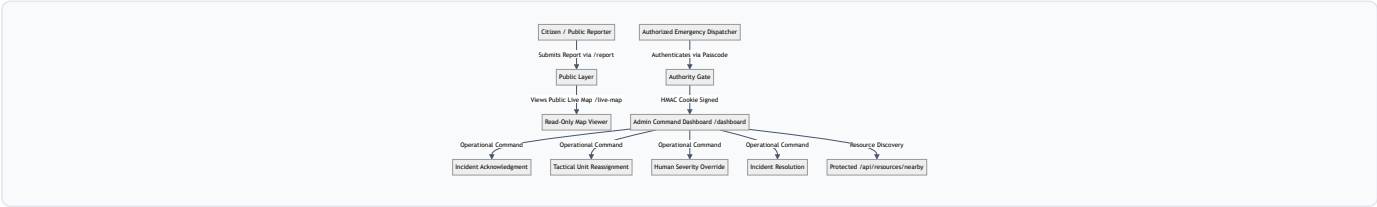
1. Provide zero-hallucination emergency resource recommendations grounded strictly in real mapped geospatial assets.
2. Ensure sub-second mathematical confidence scoring with transparent, auditable point contributor breakdowns.
3. Deliver real-time investigation visibility via Server-Sent Events (SSE) streaming during citizen report submission.
4. Guarantee fail-safe resilience via multi-tier fallbacks across Gemini AI, Overpass API, Nominatim geocoding, and Cloud Firestore storage.
5. Provide strict server-side cryptographic authentication for emergency responder command access.

3.3 Non-Goals VERIFIED IN CODE

1. **No Autonomous Dispatch:** The system will never dispatch real municipal emergency units without explicit human dispatcher confirmation.
2. **No Real-Time Unit Availability Tracking:** The system does not claim to know live paramedic staffing, current truck call load, or open hospital ICU beds (explicitly acknowledged as an unmapped variable).
3. **No Direct Turn-by-Turn Routing:** The platform computes great-circle Haversine physical proximity rather than vehicular travel-time road networks.
4. **No Direct Telephony PBX Integration:** Telephony actions leverage standard OS-level URI protocols (`tel:<phone>`) rather than integrated WebRTC SIP trunks.

3. Stakeholders & User Roles

3.1 Role Hierarchy VERIFIED IN CODE

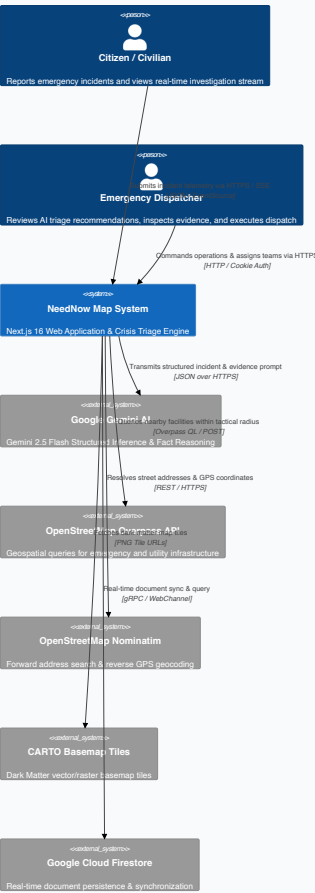


3.2 Role Matrix VERIFIED IN CODE

Role	Access Level	Authentication Method	Available Capabilities
Citizen Reporter	Public	None required	Access <code>/</code> , <code>/report</code> , <code>/live-map</code> (read-only), trigger <code>/api/triage/stream</code> , <code>/api/analyze</code> , submit incidents to <code>/api/incidents</code> .
Public Map Viewer	Public	None required	Access <code>/live-map</code> to inspect active public incident markers and summary details in read-only mode.
Emergency Dispatcher	Authorized	Passcode (<code>PASSCODE</code>) → Signed <code>authority_session</code> HMAC cookie	Full access to <code>/dashboard</code> , <code>/api/resources/nearby</code> , PATCH operations on <code>/api/incidents/[id]</code> , team assignment, human override, status resolution, and audit logs.
System Administrator	Server Operator	Direct server environment / Cloud Console	Configure environment secrets (<code>GEMINI_API_KEY</code> , <code>PASSCODE</code> , <code>MAP_KEY</code> , <code>FIREBASE_*</code>), monitor server logs, and configure Overpass API mirrors.

4. System Context & Operational Boundaries

System Context Diagram – NeedNow Map Platform



5. Complete Repository & Module Architecture

5.1 Directory Tree VERIFIED IN CODE

```
/Users/srujan/Beast/Coding/web/gdg
├── AGENTS.md                # Next.js agent operational rules
├── CLAUDE.md                # Architecture guidelines
├── README.md               # Repository overview & setup
├── SRS.md                  # Software Requirements Specification
├── eslint.config.mjs       # ESLint 9 configuration
├── next.config.ts          # Next.js 16 configuration
├── package.json            # Dependencies, scripts & engine metadata
├── package-lock.json       # Deterministic dependency lockfile
├── postcss.config.mjs      # Tailwind CSS v4 PostCSS integration
├── tsconfig.json           # TypeScript 5 compiler options & path aliases
├── .env.example            # Environment template with secret keys
├── .env.local              # Local environment configuration
├──
├── public/                 # Static web assets
│   ├── file.svg
│   ├── globe.svg
│   ├── next.svg
│   ├── vercel.svg
│   └── window.svg
├──
├── scripts/                # Automated test & acceptance scripts
│   └── test_phase2.mjs     # 12-suite Phase 2 automated test harness
├──
├── src/                    # Application Source Code
│   ├── app/               # Next.js App Router (Pages, Layouts & API)
│   │   ├── favicon.ico    # Application browser icon
│   │   ├── globals.css    # Tailwind v4 theme & CSS variable tokens
│   │   ├── layout.tsx     # Root layout with Geist font & AuthorityProvider
│   │   └── page.tsx       # Public landing page & operational matrix
│   │
│   ├── dashboard/         # Dispatcher Command Center
│   │   ├── layout.tsx     # AuthorityGate authentication barrier
│   │   └── page.tsx       # Command Center HUD, timeline replay & split view
│   │
│   ├── incident/[id]/     # Direct Incident Deep Link
│   │   └── page.tsx       # Server-rendered individual incident telemetry
│   │
│   ├── live-map/          # Fullscreen Tactical Map Array
│   │   └── page.tsx       # Realtime map tracker with read-only inspector
│   │
│   ├── report/            # Citizen Intake & Investigation Flow
│   │   └── page.tsx       # Report intake, SSE visualizer & triage submission
│   │
│   ├── api/               # REST & Streaming Server-Side API Handlers
│   │   ├── analyze/       # POST: Legacy REST incident analysis endpoint
│   │   │   └── route.ts
│   │   ├── auth/         # GET/POST/DELETE: Passcode verification & session management
│   │   │   └── authority/
│   │   │       └── route.ts
│   │   ├── incidents/     # GET/POST: Incident collection retrieval & creation
│   │   │   ├── route.ts
│   │   │   └── [id]/
│   │   │       └── route.ts
│   │   ├── map-config/    # PATCH: Protected incident update & team assignment
│   │   │   ├── route.ts
│   │   │   └── route.ts
│   │   ├── resources/     # GET: Dynamic CARTO basemap tile URL resolver
│   │   │   └── nearby/
│   │   │       └── route.ts
│   │   ├── triage/        # GET: Protected live OSM capability discovery query
│   │   │   ├── route.ts
│   │   │   └── route.ts
│   │   ├── triage/        # POST: Synchronous evidence-based triage handler
│   │   │   ├── route.ts
│   │   │   └── route.ts
│   │   └── stream/        # POST: SSE ReadableStream live investigation engine
│   │       └── route.ts
│   │
│   ├── components/        # React UI Components
│   │   ├── layout/        # Shell & Navigation Layout Components
│   │   │   ├── AppShell.tsx # Global shell with Sidebar, MobileNav & Banner
│   │   │   ├── MobileNav.tsx # Mobile navigation header & slide-out drawer
│   │   │   └── Sidebar.tsx   # Desktop navigation sidebar & authority status
│   │   └── shared/        # Reusable Presentation & Tactical UI Components
│   │       ├── AuthorityGate.tsx # Passcode modal barrier for restricted routes
│   │       ├── EmptyState.tsx   # Filter empty state placeholder
│   │       ├── FirebaseWarningBanner.tsx # Non-blocking dev banner for missing Firestore
│   │       ├── IncidentCard.tsx  # Queue card with resource pills & confidence badge
│   │       ├── IncidentDetailPanel.tsx # 5-section Command Console & Dispatch Action HUD
│   │       ├── LiveMap.tsx      # Leaflet map with layer toggles & relationship vectors
│   │       ├── LoadingState.tsx  # Standardized loading indicator
│   │       └── MapWrapper.tsx   # SSR-safe dynamic wrapper for LiveMap
```

├─ MiniMap.tsx	# Compact Leaflet map for citizen intake
├─ SeverityBadge.tsx	# Red / Amber / Green severity indicator pill
├─ StatCard.tsx	# Operational metric card
└─ Toaster.tsx	# Global notification toaster (CustomEvent powered)
├─ contexts/	# React Context Providers
└─ AuthorityContext.tsx	# Client-side authority state & session management
├─ data/	# Seed & Static Application Constants
└─ constants.ts	# Pre-configured MOCK_INCIDENTS & application constants
├─ hooks/	# Custom React Hooks
├─ useIncidents.ts	# Firestore singleton listener & REST polling hook
└─ useMapConfig.ts	# Shared CARTO basemap tile configuration hook
├─ lib/	# Server & Client Utility Libraries
├─ ai.ts	# AI module interface wrapper
├─ auth.ts	# Cryptographic HMAC signing & timing-safe verification
├─ confidenceCalculator.ts	# Pure deterministic mathematical confidence engine
├─ firebase.ts	# Cloud Firestore client initialization
├─ geospatial.ts	# Overpass queries, Haversine math & contact parser
├─ investigationEngine.ts	# Multi-stage investigation pipeline & Gemini reasoner
├─ memoryStore.ts	# In-memory circular fallback incident storage
├─ resourceRanker.ts	# Capability registry & deterministic resource ranker
└─ utils.ts	# Tailwind CSS class merging ('cn')
├─ types/	# TypeScript Type & Interface Definitions
├─ incident.ts	# Incident domain model, categories & lifecycle types
└─ investigation.ts	# Capability registry, evidence & confidence schemas

6. File-by-File Technical Reference

6.1 Configuration & Root Environment Files

package.json VERIFIED IN CONFIG

- File Path:** /Users/srujan/Beast/Coding/web/gdg/package.json
- Purpose:** Declares package dependencies, build scripts, and runtime environment versions.
- Responsibility:** Defines runtime libraries (@google/genai , leaflet , firebase , framer-motion , lucide-react , next , react , react-leaflet , tailwind-merge) and dev dependencies (tailwindcss , @tailwindcss/postcss , typescript , eslint).
- Category:** Configuration.
- Server/Client:** Build-time.

tsconfig.json VERIFIED IN CONFIG

- File Path:** /Users/srujan/Beast/Coding/web/gdg/tsconfig.json
- Purpose:** Configures TypeScript compiler options.
- Responsibility:** Sets target: ES2017 , module: esnext , strict: true , jsx: react-jsx , and path alias mapping @/* → ./src/* .
- Category:** Configuration.

next.config.ts VERIFIED IN CONFIG

- File Path:** /Users/srujan/Beast/Coding/web/gdg/next.config.ts
- Purpose:** Configures Next.js App Router and Turbopack compiler.
- Responsibility:** Standard Next.js config export satisfying Next.js 16 type definitions.
- Category:** Configuration.

postcss.config.mjs VERIFIED IN CONFIG

- File Path:** /Users/srujan/Beast/Coding/web/gdg/postcss.config.mjs
- Purpose:** Configures PostCSS pipeline with @tailwindcss/postcss for Tailwind CSS v4.
- Category:** Configuration.

.env.example VERIFIED IN CONFIG

- File Path:** /Users/srujan/Beast/Coding/web/gdg/.env.example
- Purpose:** Template for required environment variables.
- Exports/Keys Declared:** NEXT_PUBLIC_FIREBASE_API_KEY , NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN , NEXT_PUBLIC_FIREBASE_PROJECT_ID , NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET , NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID , NEXT_PUBLIC_FIREBASE_APP_ID , GEMINI_API_KEY , PASSCODE , MAP_KEY .

- **Security:** Template contains mock/redacted values.
- **Category:** Configuration.

6.2 Data Models & TypeScript Definitions (`src/types/`)

`src/types/incident.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/types/incident.ts`
- **Purpose:** Core domain model definitions for incidents, categories, statuses, urgency, and operational timelines.
- **Exports:**
 - `SeverityZone` : "red" | "amber" | "green"
 - `IncidentStatus` : "new" | "reviewed" | "assigned" | "resolved"
 - `IncidentCategory` : 16 categories (`flood`, `fire`, `road_blockage`, `injury`, `power_outage`, `supply_shortage`, `trapped_people`, `medical_emergency`, `hazard`, `other`, `water_leak`, `water_rescue`, `electrical_hazard`, `gas_leak`, `structural_collapse`, `industrial_hazard`).
 - `UrgencyLevel` : "immediate" | "high" | "moderate" | "low"
 - `IncidentTimelineEvent` : Structure for audit trail timeline events (`timestamp`, `stage`, `title`, `description`, `actor`).
 - `Incident` : Comprehensive interface encompassing all incident fields, telemetry, confidence breakdown, attached resource snapshots, and override states.
- **Dependencies:** Imports `ConfidenceBreakdown`, `ContradictionRecord`, `RankedOperationalResource` from `./investigation`.
- **Consumers:** Consumed by nearly all UI components, hooks, API routes, and ranking engines.
- **Category:** Core Types.

`src/types/investigation.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/types/investigation.ts`
- **Purpose:** Schema definitions for resource capabilities, investigation steps, geospatial evidence, resource ranking, and SSE streaming events.
- **Exports:**
 - `LocationQuality` : "exact_gps" | "resolved_address" | "approximate_city" | "unresolved"
 - `ResourceCapability` : 13 canonical capabilities (`fire_suppression`, `heavy_extrication_usar`, `hazmat_containment`, `trauma_care`, `ems_transport`, `swift_water_rescue`, `traffic_perimeter`, `evacuation_support`, `power_grid_isolation`, `gas_grid_isolation`, `water_grid_isolation`, `public_works_clearing`, `critical_facility_backup`).
 - `ResourceType` : 15 entity types (`fire_station`, `hospital`, `ambulance_station`, `police_station`, `rescue_station`, `water_rescue_station`, `power_substation`, `power_transformer`, `gas_pipeline`, `gasometer`, `water_works`, `water_utility`, `public_works_depot`, `hospital_facility`, `emergency_resource`).
 - `InvestigationStepType` & `InvestigationStep` : Models investigation progress (`status`, `title`, `message`, `metadata`).
 - `TriageEvidenceType` & `TriageEvidence` : Normalized OpenStreetMap evidence model with field-level contact provenance.
 - `RankedOperationalResource` : Rich operational resource model with relevance score, rank, usability flags, and recommendation reasons.
 - `ConfidenceFactor`, `ConfidenceBreakdown`, `ConfidenceEngineInput` : Input and output models for deterministic confidence calculation.
 - `ContradictionRecord` : Model for discrepancies between structured form fields and narrative text.
 - `EvidenceBasedTriageResponse` : Complete output payload of the investigation engine.
 - `InvestigationStreamEvent` : Discriminated union of all 8 SSE event types (`step_update`, `search_started`, `evidence_found`, `resource_found`, `search_completed`, `resource_ranked`, `quality_assessed`, `triage_complete`, `investigation_error`).
- **Category:** Core Types.

6.3 Core Business Logic & Libraries (`src/lib/`)

`src/lib/auth.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/auth.ts`
- **Purpose:** Server-side cryptographic authentication and authorization helpers.
- **Exports:**
 - `AUTHORITY_COOKIE_NAME` : Constant (`"authority_session"`).
 - `verifyPasscode(inputPasscode: string): boolean` : Uses `crypto.timingSafeEqual` to verify the emergency passcode against `process.env.PASSCODE || "demo123"`.
 - `createSessionToken(): string` : Creates a signed HMAC-SHA256 token in the format `<timestamp>.<signature>` with a 7-day TTL.
 - `verifySessionToken(token: string): boolean` : Validates token signature and checks timestamp expiration (max age 604,800 seconds).
 - `isAuthorized(): Promise<boolean>` : Asynchronously reads the incoming request cookies and verifies the session token.
- **Security:** Uses timing-safe byte comparison to eliminate side-channel timing attacks.

- **Server/Client:** Strictly server-side (`next/headers` , `crypto`).
- **Consumers:** `src/app/api/auth/authority/route.ts` , `src/app/api/resources/nearby/route.ts` , `src/app/api/incidents/[id]/route.ts` .
- **Category:** Core Security.

`src/lib/confidenceCalculator.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/confidenceCalculator.ts`
- **Purpose:** Pure deterministic mathematical calculation of multi-dimensional confidence scores.
- **Exports:**
 - `calculateDeterministicConfidence(input: ConfidenceEngineInput): ConfidenceBreakdown` : Computes Location (L), Classification (C), Severity (S), and Evidence (E) dimension scores, factors in positive/negative points, applies discrepancy deductions, and yields a clamped overall score `[15, 99]`.
- **Deterministic Math:** Does not call Gemini; 100% computed via server-side TS logic.
- **Consumers:** `src/lib/investigationEngine.ts` .
- **Category:** Core Intelligence.

`src/lib/resourceRanker.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/resourceRanker.ts`
- **Purpose:** Formal Canonical Capability Registry declaration and deterministic operational resource ranking engine.
- **Exports:**
 - `CANONICAL_CAPABILITY_REGISTRY` : Formal registry defining all 13 capabilities, their entity kinds (`emergency_response` vs `contextual_infrastructure`), default search radii, primary/fallback OSM types, and fallback notes.
 - `calculateProximityScore(distanceKm: number): number` : Computes exponential decay proximity score $P(d) = 100 \times \exp(-0.15 \times d)$ clamped to `[5, 100]`.
 - `rankOperationalResources(evidenceList: TriageEvidence[], requiredCapabilities: ResourceCapability[])` : `RankedOperationalResource[]` : Evaluates relevance score ($0.50 \times \text{CapMatch} + 0.40 \times P(d) + 0.10 \times \text{Context}$), tags usability flags, extracts contact telemetry, assigns ranks, sets `isPrimaryRecommendation` , and builds recommendation rationale strings.
- **Consumers:** `src/lib/investigationEngine.ts` , `src/app/api/resources/nearby/route.ts` .
- **Category:** Core Intelligence.

`src/lib/geospatial.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/geospatial.ts`
- **Purpose:** Geospatial queries against OpenStreetMap Overpass API mirrors, Haversine distance calculation, contact extraction, and in-memory spatial caching.
- **Exports:**
 - `calculateHaversineDistanceKm(lat1, lon1, lat2, lon2): number` : High-precision great-circle distance in kilometers rounded to 1 decimal.
 - `searchNearbyCapabilities(lat, lng, capabilities, radiusKm): Promise<{ evidence: TriageEvidence[]; success: boolean; error?: string }>` : Builds capability-specific Overpass QL query, searches up to 4 mirror endpoints with 8.0s timeouts, checks 5-minute spatial cache, normalizes results, extracts contact provenance (`OSM:phone` , `OSM:website` , `OSM:addr:full`), and deduplicates by coordinates.
 - `searchNearbyInfrastructure(...)` : Backward-compatible wrapper for legacy callers.
- **Consumers:** `src/lib/investigationEngine.ts` , `src/app/api/resources/nearby/route.ts` .
- **Category:** Core Geospatial.

`src/lib/investigationEngine.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/investigationEngine.ts`
- **Purpose:** Central orchestration engine for the multi-stage Evidence-Based Investigation Pipeline.
- **Exports:**
 - `IncidentInputPayload` : Input payload interface.
 - `runInvestigationPipeline(payload, emitEvent?): Promise<EvidenceBasedTriageResponse>` : Executes all 8 investigation stages:
 1. Incident parsing & location quality evaluation.
 2. Incident-aware capability planning (positive & negative scoping).
 3. Overpass geospatial capability search & SSE event emission.
 4. Deterministic resource ranking (`rankOperationalResources`).
 5. Evidence quality & missing telemetry assessment.
 6. Gemini 2.5 Flash evidence reasoning (or rule-based fallback).
 7. Deterministic confidence computation (`calculateDeterministicConfidence`).
 8. Final result assembly and completion event broadcast.
- **Dependencies:** `@google/genai` , `src/lib/geospatial.ts` , `src/lib/confidenceCalculator.ts` , `src/lib/resourceRanker.ts` .

- **Consumers:** `src/app/api/triage/stream/route.ts`, `src/app/api/triage/route.ts`, `src/app/api/analyze/route.ts`, `src/lib/ai.ts`.
- **Category:** Core Orchestration.

`src/lib/ai.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/ai.ts`
- **Purpose:** Backward-compatible wrapper exposing `triageIncident()` routing directly to `runInvestigationPipeline()`.
- **Exports:** `AITriageResponse`, `triageIncident()`.
- **Category:** Support / Compatibility.

`src/lib/firebase.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/firebase.ts`
- **Purpose:** Cloud Firestore database initialization and environment variable validation.
- **Exports:**
 - `hasFirebaseConfig`: Boolean flag verifying that all 6 `NEXT_PUBLIC_FIREBASE_*` variables are non-empty.
 - `db`: Initialized Firestore instance (or `null` if unconfigured/failed).
- **Side Effects:** Emits developer-friendly console warning when Firebase environment variables are missing.
- **Category:** Persistence Configuration.

`src/lib/memoryStore.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/memoryStore.ts`
- **Purpose:** In-memory fallback data store for local development and Firestore offline resilience.
- **Exports:**
 - `memoryIncidents`: Array initialized with `MOCK_INCIDENTS`.
 - `updateMemoryIncident(id: string, updates: Partial<Incident>): void`: In-place update of memory incident.
 - `addMemoryIncident(incident: any): void`: Prepends new incident to memory store.
- **Consumers:** `src/app/api/incidents/route.ts`, `src/app/api/incidents/[id]/route.ts`.
- **Category:** Fallback Persistence.

`src/lib/utils.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/lib/utils.ts`
- **Purpose:** Tailwind CSS class utility.
- **Exports:** `cn(...inputs: ClassValue[]): string` combining `clsx` and `tailwind-merge`.
- **Category:** Utility.

6.4 Static Data & Constants (`src/data/`)

`src/data/constants.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/data/constants.ts`
- **Purpose:** Provides static mock incidents and default application configuration.
- **Exports:**
 - `MOCK_INCIDENTS`: Array of 2 complete mock incidents (`inc-blr-01`: Commercial Structural Fire in MG Road; `inc-blr-02`: Municipal Water Main Rupture in Indiranagar) containing full confidence breakdowns, mapped resources, and timeline histories.
 - `APP_CONFIG`: Application name and version constant (`"0.3.0"`).
- **Category:** Static Data.

6.5 Contexts & Client State (`src/context/`)

`src/context/AuthorityContext.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/context/AuthorityContext.tsx`
- **Purpose:** Client-side React context for tracking dispatcher authorization status and executing unlock/lock actions.
- **Exports:** `AuthorityProvider`, `useAuthority()`.
- **Side Effects:** Polls `GET /api/auth/authority` on mount to establish session state without hydration flickering.
- **Category:** Client State.

6.6 Custom React Hooks (`src/hooks/`)

`src/hooks/useIncidents.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/hooks/useIncidents.ts`
- **Purpose:** Real-time incident telemetry hook utilizing a singleton Firestore `onSnapshot` listener with reference counting and 1-second debounce teardown, falling back to 4-second REST polling against `/api/incidents`.
- **Exports:** `useIncidents(): { incidentsSource: Incident[]; loadingDb: boolean }`.
- **Consumers:** `src/app/dashboard/page.tsx`, `src/app/live-map/page.tsx`.
- **Category:** Custom Hook / Telemetry.

`src/hooks/useMapConfig.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/hooks/useMapConfig.ts`
- **Purpose:** Shared basemap configuration hook caching the CARTO tile URL resolved from `/api/map-config` across all map instances.
- **Exports:** `useMapConfig(): { tileUrl: string }`.
- **Consumers:** `src/components/shared/LiveMap.tsx`, `src/components/shared/MiniMap.tsx`.
- **Category:** Custom Hook / Geospatial.

6.7 UI & Layout Components (`src/components/`)

`src/components/layout/AppShell.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/layout/AppShell.tsx`
- **Purpose:** Root application shell wrapping pages with Sidebar, MobileNav, FirebaseWarningBanner, and Toaster.
- **Category:** UI Layout.

`src/components/layout/Sidebar.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/layout/Sidebar.tsx`
- **Purpose:** Desktop navigation sidebar displaying navigation links, authority badge, and one-click session lock control.
- **Category:** UI Layout.

`src/components/layout/MobileNav.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/layout/MobileNav.tsx`
- **Purpose:** Responsive navigation bar with hamburger menu drawer for mobile viewports.
- **Category:** UI Layout.

`src/components/shared/AuthorityGate.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/AuthorityGate.tsx`
- **Purpose:** Fullscreen modal barrier intercepting unauthorized visitors on restricted routes (`/dashboard`), prompting for `PASSCODE` and calling `unlock()`.
- **Category:** UI Security.

`src/components/shared/LiveMap.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/LiveMap.tsx`
- **Purpose:** Client-side Leaflet map rendering incident markers, resource markers, active search radii, single-vector tactical relationship lines to top recommendations, layer toggles ( Fire ,  Medical ,  Utilities ,  Police), and custom dark popups.
- **Category:** UI Geospatial.

`src/components/shared/MapWrapper.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/MapWrapper.tsx`
- **Purpose:** Dynamic SSR-safe wrapper for `LiveMap.tsx` with `ssr: false` to bypass Leaflet `window is not defined` exceptions during Next.js server rendering.
- **Category:** UI Wrapper.

`src/components/shared/MiniMap.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/MiniMap.tsx`
- **Purpose:** Compact Leaflet map component used on `/report` to display incident coordinates, animated scanning search circles, and discovered Overpass evidence markers.
- **Category:** UI Geospatial.

`src/components/shared/IncidentCard.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/IncidentCard.tsx`
- **Purpose:** Incident queue item card displaying zone border glow, confidence badge, category, summary, resource pill counts, casualties, and selection highlight.
- **Category:** UI Component.

`src/components/shared/IncidentDetailPanel.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/IncidentDetailPanel.tsx`
- **Purpose:** 5-section Incident Command Console drawer providing situation telemetry, AI assessment matrix, expandable confidence drawer, ranked resources panel with direct calling (`tel:`) and portal actions, tactical directives, assignment history, and human dispatcher controls.
- **Category:** UI Component.

`src/components/shared/SeverityBadge.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/SeverityBadge.tsx`
- **Purpose:** Renders styled badges for Critical (Red), Caution (Amber), and Stable (Green) zones.
- **Category:** UI Component.

`src/components/shared/StatCard.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/StatCard.tsx`
- **Purpose:** Metric card component displaying numerical values, titles, icons, and descriptions in the command center HUD.
- **Category:** UI Component.

`src/components/shared/Toaster.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/Toaster.tsx`
- **Purpose:** Notification toast manager triggered via `document.dispatchEvent(new CustomEvent('global-toast', ...))` rendering animated Framer Motion toast cards.
- **Category:** UI Component.

`src/components/shared/FirebaseWarningBanner.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/FirebaseWarningBanner.tsx`
- **Purpose:** Non-blocking warning banner shown during development if Firebase credentials are missing.
- **Category:** UI Component.

`src/components/shared/EmptyState.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/EmptyState.tsx`
- **Purpose:** Placeholder displayed when incident filter queries yield 0 results.
- **Category:** UI Component.

`src/components/shared/LoadingState.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/components/shared/LoadingState.tsx`
- **Purpose:** Standardized spinner and loading message container.
- **Category:** UI Component.

6.8 Pages & Routes (`src/app/`)

`src/app/page.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/page.tsx`
- **Purpose:** Public landing page highlighting system value propositions, 4-stage operational flow (Ingest → Assess → Project → Deploy), and quick-launch action buttons.
- **Category:** Page.

`src/app/dashboard/page.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/dashboard/page.tsx`
- **Purpose:** Command Center Dashboard featuring situational KPI HUD, 24-hour timeline history replay slider, interactive tactical filters, 60/40 map-and-queue split view, and incident selection.
- **Category:** Page.

`src/app/dashboard/layout.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/dashboard/layout.tsx`
- **Purpose:** Wraps the dashboard in `AuthorityGate` to enforce passcode authentication.
- **Category:** Page Layout.

`src/app/live-map/page.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/live-map/page.tsx`
- **Purpose:** Fullscreen geospatial tracking array displaying active incidents and read-only telemetry drawers.
- **Category:** Page.

`src/app/report/page.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/report/page.tsx`
- **Purpose:** Citizen incident intake page featuring address search autocomplete, GPS capture, live mini-map, SSE investigation stream visualizer, evidence cards, confidence breakdown, human override, and dispatch submission.
- **Category:** Page.

`src/app/incident/[id]/page.tsx` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/incident/[id]/page.tsx`
- **Purpose:** Server-side rendered direct permalink page for individual incident inspection from Firestore.
- **Category:** Page.

6.9 API Route Handlers (`src/app/api/`)

`src/app/api/auth/authority/route.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/api/auth/authority/route.ts`
- **Handlers:** `GET` , `POST` , `DELETE` .
- **Purpose:** Manages dispatcher passcode verification, HMAC cookie creation, session verification, and logout.
- **Category:** API Handler.

`src/app/api/resources/nearby/route.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/api/resources/nearby/route.ts`
- **Handlers:** `GET` .
- **Purpose:** Protected endpoint querying live OpenStreetMap Overpass resources around given coordinates and returning deterministically ranked resources. Requires valid `authority_session` cookie.
- **Category:** API Handler.

`src/app/api/map-config/route.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/api/map-config/route.ts`
- **Handlers:** `GET` .
- **Purpose:** Dynamically constructs the CARTO Dark Matter basemap tile URL using server-side `process.env.MAP_KEY` and sets 1-hour HTTP caching headers.
- **Category:** API Handler.

`src/app/api/triage/stream/route.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/api/triage/stream/route.ts`
- **Handlers:** `POST` .
- **Purpose:** SSE streaming route executing `runInvestigationPipeline()` and emitting real-time investigation events via a `ReadableStream` .
- **Category:** API Handler.

`src/app/api/triage/route.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/api/triage/route.ts`
- **Handlers:** `POST` .
- **Purpose:** Synchronous REST endpoint executing the full investigation pipeline and returning `EvidenceBasedTriageResponse` .
- **Category:** API Handler.

`src/app/api/analyze/route.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/api/analyze/route.ts`

- **Handlers:** `POST` .
- **Purpose:** Legacy backward-compatible REST analysis route routing to `runInvestigationPipeline()` .
- **Category:** API Handler.

`src/app/api/incidents/route.ts` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/api/incidents/route.ts`
- **Handlers:** `GET` , `POST` .
- **Purpose:** Ingests new incidents into Cloud Firestore (and `memoryStore.ts`), and retrieves all persisted incidents sorted by `updatedAt` descending.
- **Category:** API Handler.

`src/app/api/incidents/[id]/route.ts` VERIFIED IN CODE

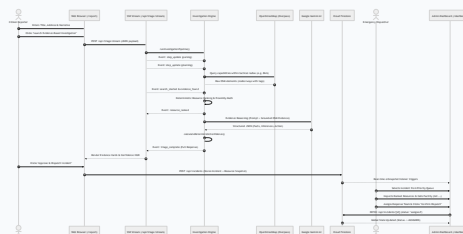
- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/src/app/api/incidents/[id]/route.ts`
- **Handlers:** `PATCH` .
- **Purpose:** Protected endpoint for dispatchers to update incident status, assign response teams, update timelines, or record human overrides. Requires `isAuthorized()` .
- **Category:** API Handler.

6.10 Test Suites (`scripts/`)

`scripts/test_phase2.mjs` VERIFIED IN CODE

- **File Path:** `/Users/srujan/Beast/Coding/web/gdg/scripts/test_phase2.mjs`
- **Purpose:** Automated acceptance test suite validating 12 distinct functional scenarios:
 1. Electrical Fire + 6 Trapped People (validates fire suppression, power isolation, trauma care, negative check on water rescue, confidence ≥ 75).
 2. Municipal Water Main Rupture (validates water grid isolation, public works, negative check on water rescue and trauma care).
 3. Regional River Flood (validates swift water rescue, fire suppression, evacuation support).
 4. Medical Cardiac Emergency (validates trauma care, EMS transport, negative check on heavy USAR).
 5. Road Blockage by Fallen Tree (validates public works clearing, traffic perimeter, negative check on trauma care).
 6. Power Grid Outage (Standard Grid without hospital context).
 7. Power Grid Outage with Hospital ICU generator fuel context (validates critical facility backup).
 8. Gas Main Rupture (validates hazmat containment, gas grid isolation, evacuation support).
 9. Structural Slab Collapse (validates heavy extrication USAR, trauma care).
 10. Contact Provenance & Zero Hallucinations (validates `OSM:phone` provenance and `hasDirectPhone`).
 11. Protected Endpoint Security (validates 401 on unauthenticated `/api/resources/nearby` and 200 after passcode authentication).
 12. Incident Lifecycle & Snapshot Persistence (validates `POST /api/incidents` and `PATCH /api/incidents/:id`).
- **Category:** Automated Testing.

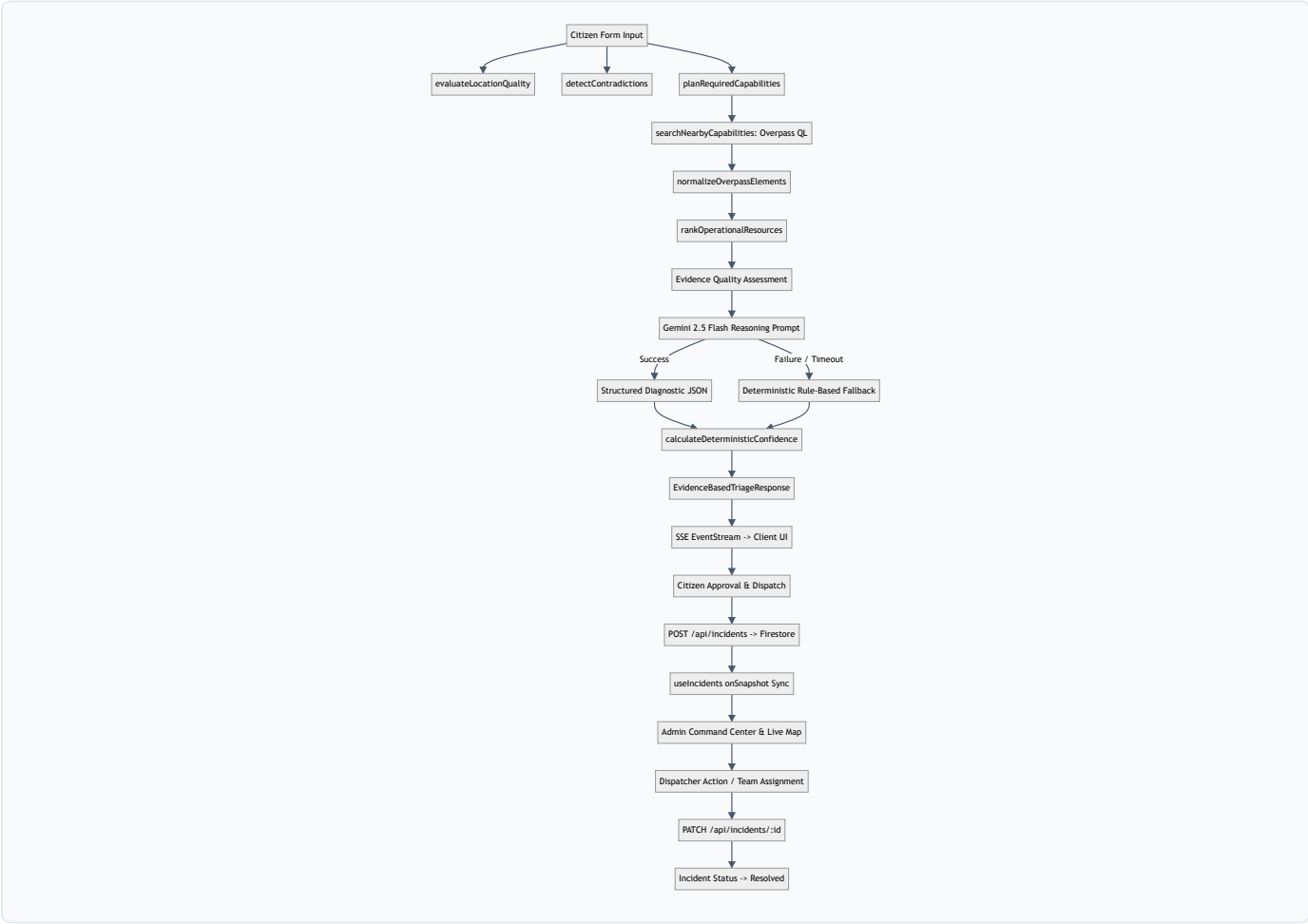
7. End-to-End User Journeys



8. System Architecture & Multi-Tier Topology



9. End-to-End Data Lifecycle & Flow Diagrams



10. Incident Data Model Specification

10.1 Schema Definition VERIFIED IN CODE

The canonical `Incident` interface (`src/types/incident.ts`) governs all incident lifecycle states, attached resource snapshots, and audit records.

```

export interface Incident {
  // Identity & Core Descriptive Attributes
  id: string; // Unique identifier (e.g. "inc-1724912345678" or Firestore doc ID)
  title: string; // Short incident title
  description: string; // Full situation narrative
  category: IncidentCategory; // One of 16 incident categories

  // Tactical Severity & Urgency
  severityScore: number; // 0-100 integer
  zone: SeverityZone; // "red" | "amber" | "green"
  urgency: UrgencyLevel; // "immediate" | "high" | "moderate" | "low"

  // Strategic Synthesis
  needs: string[]; // Required operational resources/equipment
  summary: string; // High-level executive situational summary
  bestNextAction: string; // Immediate tactical command directive
  confidence: number; // 0-100 deterministic overall confidence score

  // Lifecycle & Status
  status: IncidentStatus; // "new" | "reviewed" | "assigned" | "resolved"

  // Geospatial Coordinates & Origin
  location: string; // Descriptive address or placename
  lat: number; // WGS-84 Latitude coordinate
  lng: number; // WGS-84 Longitude coordinate

  // Provenance & Media
  reportedBy: string; // Reporting source ("Citizen Alert", "Civic Sensor Array", etc.)
  photoURL?: string; // Optional attachment URL

  // Timestamps (ISO 8601 Strings)
  createdAt: string; // Creation timestamp
  updatedAt: string; // Last updated timestamp
  resolvedAt?: string; // Optional resolution timestamp

  // Operational Telemetry
  peopleAffected: number; // Number of people impacted (operationally considered)
  reportedPeopleAffected?: number; // Raw number of people reported on intake form
  responseTeam?: string; // Currently assigned response team
  recommendedTeam?: string; // AI-recommended response team
  assignmentHistory?: Array<{ // Audit trail of team assignments
    team: string;
    source: "ai" | "manual";
    timestamp: string;
  }>;
  reasoning?: string; // AI diagnostic reasoning narrative
  lastUpdatedText?: string; // Human-readable relative time ("12m ago")

  // Audit & Epistemic Telemetry
  confidenceBreakdown?: ConfidenceBreakdown; // 4-dimension confidence breakdown & factor list
  evidenceCount?: number; // Total count of verified geospatial assets discovered
  missingEvidence?: string[]; // Known limitations & unmonitored variables
  contradictions?: ContradictionRecord[]; // Data discrepancies between form and narrative
  isOverridden?: boolean; // True if dispatcher modified AI recommendations
  originalAiAssessment?: { // Preserved baseline AI assessment
    severityScore: number;
    zone: SeverityZone;
    urgency: UrgencyLevel;
    recommendedTeam?: string;
  };

  // Attached Snapshots
  resources?: RankedOperationalResource[]; // Mapped operational resources attached at triage
  timeline?: IncidentTimelineEvent[]; // Full lifecycle timeline event history
  capabilitiesEvaluated?: string[]; // List of capabilities evaluated during investigation
}

```

11. AI Triage & Investigation Data Models

11.1 EvidenceBasedTriageResponse Schema VERIFIED IN CODE

Defined in `src/types/investigation.ts`, this represents the complete structured output produced by `runInvestigationPipeline()`:

```
export interface EvidenceBasedTriageResponse {
  category: IncidentCategory;
  severityScore: number;
  zone: SeverityZone;
  urgency: UrgencyLevel;
  needs: string[];
  summary: string;
  bestNextAction: string;
  confidence: number;
  confidenceBreakdown: ConfidenceBreakdown;
  reasoning: string;
  factsIdentified: string[];
  inferencesMade: string[];
  unknownsAcknowledged: string[];
  recommendedTeam: string;
  estimatedPeopleAffected: number;
  reportedPeopleAffected: number;
  evidence: TriageEvidence[];
  rankedResources: RankedOperationalResource[];
  capabilitiesEvaluated: ResourceCapability[];
  missingEvidence: string[];
  contradictions: ContradictionRecord[];
  investigationSteps: InvestigationStep[];
  locationQuality: LocationQuality;
  isDegradedMode: boolean;
  degradedReason?: string;
}
```

12. Google Gemini Multi-Stage AI Architecture

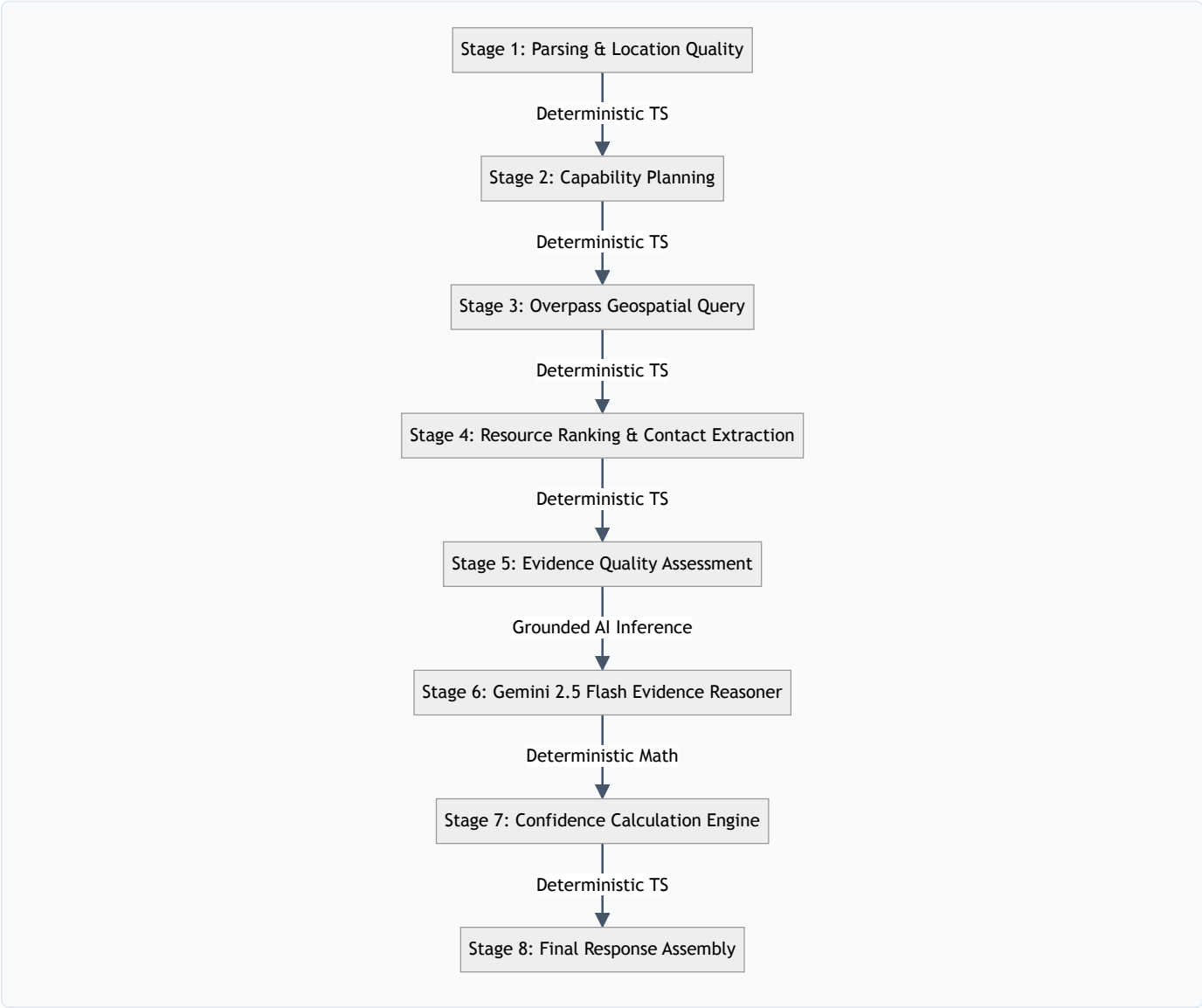
12.1 SDK & Model Configuration

VERIFIED IN CODE

- **SDK:** `@google/genai` (v1.50.1) using the official `GoogleGenAI` client instance.
- **Configured Model:** `process.env.GEMINI_MODEL || "gemini-2.5-flash"`.
- **Output Mode:** `responseMimeType: "application/json"` with strict `responseSchema` validation via `Type.OBJECT`.

12.2 Multi-Stage Pipeline Breakdown

VERIFIED IN CODE



12.3 AI vs Deterministic Division Matrix

VERIFIED IN CODE

Pipeline Subsystem	Execution Method	Responsible Module	Description
Location Precision Evaluation	Deterministic Logic	investigationEngine.ts	Evaluates telemetry source (browser → exact_gps , suggestion → resolved_address , etc.).
Contradiction Detection	Deterministic Regex & Math	investigationEngine.ts	Compares numeric peopleAffected form input against regex matches in narrative.
Capability Planning	Deterministic Rule Matrix	investigationEngine.ts	Evaluates positive and negative scoping rules across 13 canonical capabilities.
Geospatial Infrastructure Discovery	External API Query	geospatial.ts	Queries OpenStreetMap Overpass API mirrors with Haversine distance calculations.
Contact Telemetry Extraction	Deterministic Tag Parser	geospatial.ts	Extracts real OSM tags (phone , website , addr:*) with field provenance (OSM:phone).
Resource Ranking & Scoring	Deterministic Formula	resourceRanker.ts	Computes relevance score (0.50 × CapMatch + 0.40 × P(d) + 0.10 × Context).
Evidence Synthesis & Reasoning	Google Gemini 2.5 Flash	investigationEngine.ts	Synthesizes verified OSM evidence with narrative facts to formulate tactical directives.
Numeric Confidence Scoring	Deterministic Mathematical Formula	confidenceCalculator.ts	Weighted multi-dimensional calculation with universal and discrepancy penalties.

13. Canonical Resource Capability Registry

13.1 Canonical Capabilities Matrix VERIFIED IN CODE

The platform defines 13 canonical capabilities in `CANONICAL_CAPABILITY_REGISTRY` (`src/lib/resourceRanker.ts`):

Capability Identifier	Label	Entity Kind	Category	Default Radius	Primary OSM Types	Fallback Note / Limitation
fire_suppression	Fire Suppression & Structural Protection	emergency_response	fire	8 km	fire_station	Mapped disaster unit — structural fire suppression capability unverified.
heavy_extrication_usar	Urban Search & Rescue (USAR)	emergency_response	rescue	12 km	rescue_station	Mapped fire station — specialized USAR heavy extrication unverified in public dataset.
hazmat_containment	Hazardous Materials (Hazmat) Containment	emergency_response	fire	8 km	fire_station	Mapped fire station — specialized Hazmat containment unverified in public dataset.
trauma_care	Emergency Trauma & Surgical Care	emergency_response	medical	10 km	hospital	Mapped medical facility — comprehensive 24/7 trauma ICU capability unverified.
ems_transport	Emergency Medical Services (EMS / Ambulance)	emergency_response	medical	8 km	ambulance_station	Hospital facility mapped — dedicated standby ambulance availability unverified.
swift_water_rescue	Swift Water & Flood Rescue	emergency_response	rescue	15 km	water_rescue_station	Mapped fire station — specialized swift water rescue capability unverified.
traffic_perimeter	Traffic Diversion & Perimeter Security	emergency_response	police	8 km	police_station	Security post mapped — active traffic patrol division unverified.
evacuation_support	Area Evacuation & Shelter Management	emergency_response	police	12 km	police_station , fire_station	Civic facility mapped — dedicated evacuation management team unverified.
power_grid_isolation	Power Grid Substation & High-Voltage Isolation	contextual_infrastructure	utility	6 km	power_substation , power_transformer	Substation physical asset mapped — control room operational contact unverified.
gas_grid_isolation	Gas Pipeline & Fuel Isolation	contextual_infrastructure	hazard	8 km	gas_pipeline , gasometer	Gas infrastructure asset mapped — utility valve operator dispatch required.
water_grid_isolation	Municipal Water Utility & Valve Control	contextual_infrastructure	utility	6 km	water_works , water_utility	Municipal water asset mapped — field valve technician dispatch required.

- Cache TTL: 300,000 ms (5 minutes).

15.3 Great-Circle Haversine Distance Formula VERIFIED IN CODE

$$\Delta\text{lat} = \frac{(\text{lat}_2 - \text{lat}_1)\pi}{180}, \quad \Delta\text{lon} = \frac{(\text{lon}_2 - \text{lon}_1)\pi}{180}$$
$$a = \sin^2\left(\frac{\Delta\text{lat}}{2}\right) + \cos\left(\frac{\text{lat}_1\pi}{180}\right) \cos\left(\frac{\text{lat}_2\pi}{180}\right) \sin^2\left(\frac{\Delta\text{lon}}{2}\right)$$
$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a}), \quad d = R \cdot c \quad (R = 6371 \text{ km})$$

16. Interactive Map Architecture & Basemap Configuration

16.1 Leaflet Architecture VERIFIED IN CODE

- **Components:** `LiveMap.tsx`, `MiniMap.tsx`, and `MapWrapper.tsx` (Next.js dynamic wrapper with `ssr: false`).
- **Tile Layer Provider:** CARTO Dark Matter via `/api/map-config` (`https://{s}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}{r}.png`).
- **Visual Elements:**
 - **Incident Origin Markers:** Pulsing red/amber/green DivIcons.
 - **Tactical Search Circles:** Translucent bounded search circles (4–15 km).
 - **Resource Markers:** Icon-badged facility markers (`🚒`, `🏥`, `🚓`, `🚑`, `🚒`) with distance tags.
 - **Tactical Relationship Vectors:** Dashed colored Polyline connecting selected incidents to top-ranked primary recommendations.
 - **Layer Controls:** Client toggles for Incidents, Fire, Medical, Utilities, and Police layers.

17. CARTO Basemap Security & Configuration Model

17.1 Basemap Security Specification VERIFIED IN CODE

- **Endpoint:** `GET /api/map-config` dynamically returns `{ tileUrl, hasKey }`.
- **Environment Variable:** `MAP_KEY` (server-side secret).
- **Client Exposure Model:**

⚠ IMPORTANT REQUIREMENT

When `MAP_KEY` is provided, the tile URL passed to the browser includes `?key=${MAP_KEY}`. Because raster tile requests are made directly from the client's browser to CARTO CDN servers, **the basemap key is visible in browser network telemetry**. It is classified as a *Client-Visible Service Credential*, NOT an administrative server secret.

18. Firebase / Cloud Firestore Architecture

18.1 Initialization & Singleton Listener Architecture VERIFIED IN CODE

- **Module:** `src/lib/firebase.ts` initializes Firebase App and Firestore instances when all 6 `NEXT_PUBLIC_FIREBASE_*` variables are present.
- **Reference-Counted Singleton Subscription:** Implemented in `src/hooks/useIncidents.ts`:
 - Maintains a module-level `activeListenersCount` and `subscribers` set.
 - Shares a single `onSnapshot` query stream across multiple mounted components (`DashboardPage`, `LiveMapPage`).
 - Utilizes a 1000ms debounce teardown timeout on component unmount to prevent listener thrashing during client-side route navigation.
 - Automatically falls back to 4-second REST polling against `/api/incidents` if Firestore permissions are denied or credentials are unconfigured.

19. In-Memory Storage & Circular Fallback Architecture

19.1 Memory Fallback Specification VERIFIED IN CODE

- **Module:** `src/lib/memoryStore.ts`
- **Behavior:** Stores runtime incidents in memory initialized with `MOCK_INCIDENTS`.
- **Durability:** Ephemeral. Survives client navigation but resets upon Node.js server process restart.
- **Synchronization:** Kept in sync whenever POST or PATCH requests succeed or when Firestore operations encounter network/credential exceptions.

20. Deterministic Mathematical Confidence Engine

20.1 Mathematical Formula VERIFIED IN CODE

The confidence score is computed in `src/lib/confidenceCalculator.ts` using the following exact formulas:

$$\text{Overall Confidence} = \text{clamp}\left(15, 99, \text{round}(\text{Weighted Base} - \text{Penalty Sum})\right)$$
$$\text{Weighted Base} = (C \times 0.30) + (S \times 0.25) + (E \times 0.25) + (L \times 0.20)$$

1. Location Confidence (*L*)

- `exact_gps` : $L = 98$
- `resolved_address` : $L = 92$
- `approximate_city` : $L = 65$
- `unresolved` : $L = 35$

2. Classification Confidence (*C*)

- Baseline = 75
- Text Length > 60 $\implies +15$
- Text Length < 25 $\implies -15$
- ≥ 2 Domain Emergency Keywords $\implies +10$
- $C = \text{clamp}(30, 100, C)$

3. Severity Confidence (*S*)

- Baseline = 80
- Life Safety Indicators (no contradictions) $\implies +10$
- Dynamic Escalation Keywords $\implies +8$
- $S = \text{clamp}(40, 100, S)$

4. Evidence Corroboration (*E*)

- Baseline = 75
- All capability searches executed $\implies +15$
- ≥ 1 emergency response facility found $\implies +8$
- Partial search execution $\implies -15$
- $E = \text{clamp}(25, 100, E)$

5. Penalties (Penalty Sum)

- Universal limitation deduction: +4
- Contradiction penalty: $+6 \times \text{count}(\text{contradictions})$
- Missing critical telemetry penalty: $+\min(12, 4 \times \text{count}(\text{missingEvidence}))$

21. Deterministic Resource Ranking Engine

21.1 Relevance Score Formula VERIFIED IN CODE

$$\text{Relevance Score} = \text{round}\left((\text{Capability Match} \times 0.50) + (P(d) \times 0.40) + (\text{Context Specificity} \times 0.10)\right)$$

- Capability Match:** 100 if required capability; 70 otherwise.
- Proximity Score ($P(d)$):** $100 \times \exp(-0.15 \times d)$, clamped to $[5, 100]$. ($d \leq 0 \implies 100$).
- Context Specificity:** 100 if specialized capability verified; 75 if fallback type.

22. Geospatial Evidence & Provenance Tracking

22.1 Field-Level Contact Provenance VERIFIED IN CODE

The system maps OpenStreetMap tags directly to contact attributes with provenance tags:

- `tags["contact:phone"]` $\rightarrow$ `phoneSource: "OSM:contact:phone"`

- `tags["phone"]` → `phoneSource: "OSM:phone"`
- `tags["contact:mobile"]` → `phoneSource: "OSM:contact:mobile"`
- `tags["contact:website"]` → `websiteSource: "OSM:contact:website"`
- `tags["website"]` → `websiteSource: "OSM:website"`
- `tags["addr:full"]` → `addressSource: "OSM:addr:full"`
- `tags["addr:street"]` → `addressSource: "OSM:addr:street"`

23. Server-Sent Events (SSE) Investigation Stream Protocol

23.1 SSE Protocol Contract VERIFIED IN CODE

- **Route:** `POST /api/triage/stream`
- **Response Headers:**
 - `Content-Type: text/event-stream; charset=utf-8`
 - `Cache-Control: no-cache, no-transform`
 - `Connection: keep-alive`
 - `X-Accel-Buffering: no`
- **Frame Format:** `data: <JSON_STRING>\n\n`

23.2 Event Types & Schemas VERIFIED IN CODE

1. `step_update`: { event: "step_update", step: InvestigationStep }
2. `search_started`: { event: "search_started", searchType, capability, center, radiusKm, label }
3. `evidence_found`: { event: "evidence_found", evidence: TriageEvidence }
4. `resource_found`: { event: "resource_found", resource: RankedOperationalResource }
5. `search_completed`: { event: "search_completed", searchType, capability, itemsFound, nearestDistanceKm, source }
6. `resource_ranked`: { event: "resource_ranked", capability, topResourceId, topResourceName, nearestDistanceKm }
7. `quality_assessed`: { event: "quality_assessed", completenessScore, missingCount, contradictionCount }
8. `triage_complete`: { event: "triage_complete", result: EvidenceBasedTriageResponse }
9. `investigation_error`: { event: "investigation_error", message, fallbackResult }

24. Complete API Reference & Route Contracts

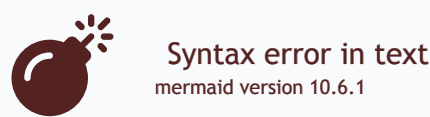
24.1 Route Catalog VERIFIED IN CODE

1. `GET /api/auth/authority`
 - **Access:** Public.
 - **Response (200):** { "isUnlocked": boolean }
2. `POST /api/auth/authority`
 - **Access:** Public.
 - **Request Body:** { "passcode": string }
 - **Response (200):** Sets `authority_session` cookie; returns { "success": true }
 - **Response (401):** { "success": false, "error": "Invalid authorization passcode." }
3. `DELETE /api/auth/authority`
 - **Access:** Public.
 - **Response (200):** Deletes cookie; returns { "success": true }
4. `GET /api/map-config`
 - **Access:** Public.
 - **Response (200):** { "tileUrl": string, "hasKey": boolean } with 1-hour cache headers.
5. `GET /api/resources/nearby`
 - **Access:** Protected (Requires `authority_session` cookie).
 - **Query Params:** `lat` (number), `lng` (number), `capability` (optional), `radiusKm` (default: 10).

- **Response (200):** { "lat": number, "lng": number, "radiusKm": number, "count": number, "resources": RankedOperationalResource[], "retrievedAt": string, "source": string }
 - **Response (401):** { "error": "Unauthorized. Authority authentication required..." }
6. `POST /api/triage/stream`
- **Access:** Public.
 - **Request Body:** `IncidentInputPayload`
 - **Response (200):** `text/event-stream` emitting SSE investigation events.
7. `POST /api/triage`
- **Access:** Public.
 - **Request Body:** `IncidentInputPayload`
 - **Response (200):** `EvidenceBasedTriageResponse`
8. `POST /api/analyze`
- **Access:** Public. Legacy alias for `/api/triage`.
9. `GET /api/incidents`
- **Access:** Public.
 - **Response (200):** `Incident[]` sorted by `updatedAt` descending.
10. `POST /api/incidents`
- **Access:** Public.
 - **Request Body:** Complete incident payload including attached resources and timeline.
 - **Response (201):** Persisted `Incident` object.
11. `PATCH /api/incidents/[id]`
- **Access: Protected** (Requires `authority_session` cookie).
 - **Request Body:** `Partial<Incident>` updates (`status`, `responseTeam`, `assignmentHistory`, etc.).
 - **Response (200):** { "success": true }
 - **Response (403):** { "error": "Unauthorized access. Admin privileges required." }

25. Authority Authentication & Cryptographic Session Model

25.1 Passcode & HMAC Session Architecture VERIFIED IN CODE



26. Security Model, Threat Analysis & Mitigations

26.1 Security Controls Matrix VERIFIED IN CODE

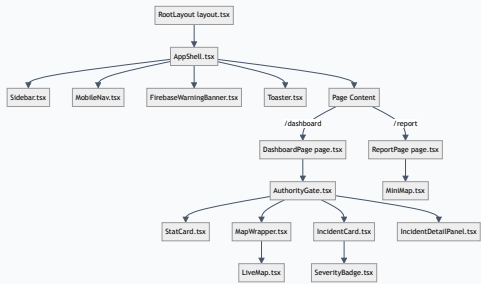
Vulnerability / Threat Area	Mitigation Strategy Implemented in Code
Passcode Timing Attacks	Implements <code>crypto.timingSafeEqual</code> in <code>src/lib/auth.ts</code> .
Session Forgery & Tampering	Server-side HMAC-SHA256 signed tokens verified against <code>ADMIN_SESSION_SECRET</code> with 7-day expiration checks.
Cross-Site Scripting (XSS)	<code>httpOnly: true</code> cookies prevent client JavaScript from reading session tokens.
Cross-Site Request Forgery (CSRF)	<code>sameSite: "lax"</code> cookie flags restrict cross-origin request transmission.
Prompt Injection / Hallucination	System prompt instructs model to distinguish Fact vs Inference vs Unknown; enforces strict JSON schema (<code>trriageReasoningSchema</code>); all contact details and coordinates come strictly from Overpass.
Basemap Key Exposure	Server routes CARTO key through <code>/api/map-config</code> . System explicitly documents key visibility in browser network calls.

27. Frontend Route Architecture

27.1 Route Table VERIFIED IN CODE

Route Path	File Path	Access Level	Primary Components	Purpose
/	src/app/page.tsx	Public	Hero, Flow Matrix, Feature Cards	Landing page and system introduction.
/dashboard	src/app/dashboard/page.tsx	Protected	AuthorityGate, LiveMap, IncidentCard, IncidentDetailPanel, StatCard	Command Center operations HUD, time replay, and triage queue.
/live-map	src/app/live-map/page.tsx	Public	MapWrapper, IncidentDetailPanel (Read-only)	Fullscreen tactical live map tracker.
/report	src/app/report/page.tsx	Public	Intake Form, MiniMap, SSE Visualizer, Evidence Cards, Override HUD	Citizen reporting and real-time investigation flow.
/incident/[id]	src/app/incident/[id]/page.tsx	Public	SeverityBadge, Location Detail Card	Server-rendered incident deep-link permalink.

28. React Component Architecture & Component Hierarchy

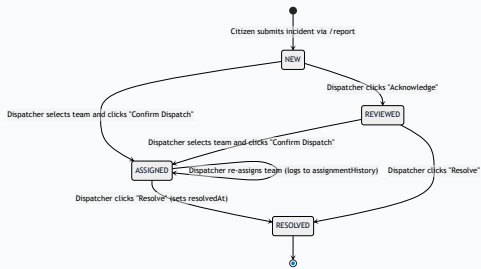


29. Custom Hooks & Client Telemetry Management

29.1 Custom Hooks Summary VERIFIED IN CODE

- `useIncidents()` (`src/hooks/useIncidents.ts`): Provides `{ incidentsSource: Incident[]; loadingDb: boolean }`. Manages singleton Firestore listener with reference counting and fallback polling.
- `useMapConfig()` (`src/hooks/useMapConfig.ts`): Provides `{ tileUrl: string }`. Fetches CARTO configuration from `/api/map-config` once and caches it across all map instances.
- `useAuthority()` (`src/contexts/AuthorityContext.tsx`): Provides `{ isUnlocked: boolean; unlock: (passcode) => Promise<boolean>; lock: () => Promise<void>; isInitialized: boolean }`.

30. Incident Lifecycle State Machine



31. Human-in-the-Loop Override & Audit Telemetry

31.1 Override Data Preservation VERIFIED IN CODE

When an operator overrides AI triage parameters:

- `isOverridden` is set to `true`.
- The original AI values are preserved in `originalAiAssessment`:

```
originalAiAssessment: {
  severityScore: number;
  zone: SeverityZone;
  urgency: UrgencyLevel;
  recommendedTeam?: string;
}
```

- Team reassignments are appended to `assignmentHistory` with `source: "manual"` or `source: "ai"`.

32. Crisis Command Center Dashboard Architecture

32.1 Dashboard Features VERIFIED IN CODE

- Top Situation Overview HUD:** Real-time counters for Active Operations, Critical Red Zone, Total Impacted Casualties, Unassigned Queue, and Avg AI Confidence.
- 24-Hour Timeline History Replay Slider:** Allows scrubbing back across 8 time steps (24h, 12h, 4h, 2h, 1h, 30m, 15m, Live) to reconstruct operational status at historical snapshots.
- Interactive Tactical Filters:** Filters by Zone (Red, Amber, Green), Status (Active, New, Acknowledged, Assigned, Resolved), Category, and Sort Order (Severity, Recency, Oldest, Confidence).
- 60/40 Split View:** Synchronized interactive map on the left and scrollable priority queue on the right.

33. Incident Command Console Specification

33.1 Five-Section Console HUD VERIFIED IN CODE

Implemented in `src/components/shared/IncidentDetailPanel.tsx`:

- Block 1: Situation Telemetry:** Zone badge, category tag, status pill, coordinates, and description.
- Block 2: AI Assessment & Epistemic Matrix:** Severity score, confidence percentage, expandable 4-dimension confidence factor drawer, and structured reasoning quote.
- Block 3: Response Intelligence & Ranked Resources:** Mapped facilities list filtered by capability tabs with direct `<a href="tel: ...">Call</a>` and `<a href="..." target="_blank">Portal</a>` links.
- Block 4: Immediate Tactical Directive:** Highlighted directive banner and casualty impact numbers.
- Block 5: Operational Timeline & Dispatch History:** Audit history of manual and AI team assignments.
- Block 6: Human Command Actions:** Team assignment dropdown, Acknowledge button, and Resolve button.

34. Live Tactical Map Tracking Array (`/live-map`)

34.1 Fullscreen Tactical Map Specification VERIFIED IN CODE

- Route:** `/live-map`
- Features:**
 - Fullscreen Leaflet map rendering active (unresolved) incidents.
 - Floating target counter HUD (`"X TARGETS IN QUEUE"`).
 - Read-only slide-out telemetry drawer on marker click.

35. Citizen Intake & Telemetry Teleportation (`/report`)

35.1 Citizen Reporting Flow VERIFIED IN CODE

- Address Autocomplete:** Debounced search against Nominatim with interactive dropdown.

- **Hardware GPS Detection:** One-click HTML5 Geolocation with reverse geocoding.
- **MiniMap Visualizer:** Displays coordinates, animated scanning circle, and live discovered assets.
- **Live SSE Investigation Feed:** Real-time progress updates across all 8 pipeline steps.
- **Dispatcher Override Mode:** Ability to adjust zone, urgency, severity, and field needs before submission.

36. Environment Variables & Configuration Matrix

36.1 Environment Variable Catalog VERIFIED IN CONFIG

Variable Name	Purpose	Visibility	Required?	Example Placeholder
NEXT_PUBLIC_FIREBASE_API_KEY	Firebase API key	Client & Server	Required for live Firestore	AIzaSy<REDACTED>
NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN	Firebase Auth domain	Client & Server	Required for live Firestore	your-project.firebaseio.com
NEXT_PUBLIC_FIREBASE_PROJECT_ID	Firestore project ID	Client & Server	Required for live Firestore	neednow-map
NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET	Firebase storage bucket	Client & Server	Required for live Firestore	your-project.firebaseiostorage.app
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID	FCM sender ID	Client & Server	Required for live Firestore	123456789012
NEXT_PUBLIC_FIREBASE_APP_ID	Firebase web app ID	Client & Server	Required for live Firestore	1:123456789012:web:abcdef
GEMINI_API_KEY	Google Gemini API key	Server-Only	Required for AI reasoning	AIzaSy<REDACTED>
GEMINI_MODEL	Gemini model name	Server-Only	Optional (Default: gemini-2.5-flash)	gemini-2.5-flash
PASSCODE	Responder access passcode	Server-Only	Required for Admin Gate	your_passcode_here
ADMIN_SESSION_SECRET	HMAC signing secret	Server-Only	Optional (Falls back to PASSCODE)	your_hmac_secret_here
MAP_KEY	CARTO basemap access key	Client-Exposed via API	Optional (Public fallback tiles used)	carto_key_here
BASE_URL	Base URL for test scripts	Script-Only	Optional (Default: http://localhost:3000)	http://localhost:3000

37. Dependency Ecosystem & Tech Stack Justification

37.1 Dependencies Matrix VERIFIED IN CONFIG

- **Core Framework:** Next.js 16.2.4 (Turbopack, App Router) · React 19.2.4 · React-DOM 19.2.4.
- **AI Inference:** @google/genai (v1.50.1) for Gemini 2.5 Flash SDK.
- **Geospatial & Mapping:** leaflet (v1.9.4), react-leaflet (v5.0.0), @types/leaflet (v1.9.21).
- **Persistence:** firebase (v12.18.0) for Cloud Firestore.
- **UI & Animations:** tailwindcss (v4.x), framer-motion (v12.38.0), lucide-react (v1.8.0), clsx (v2.1.1), tailwind-merge (v3.5.0), tailwindcss-animate (v1.0.7).

38. Build Configuration, PostCSS & SSR Constraints

38.1 SSR & Leaflet Window Constraints VERIFIED IN CODE

Leaflet references browser-only window and navigator objects during initialization. The repository resolves this cleanly:

- MapWrapper.tsx wraps LiveMap.tsx with next/dynamic and ssr: false .
- MiniMap.tsx on /report is imported via dynamic() ⇒ import(...), { ssr: false } .

39. Comprehensive Error Handling & Degraded Modes

39.1 Subsystem Failure & Degraded Matrix VERIFIED IN CODE

Failure Event	Detection Mechanism	Automated Fallback Behavior	Resulting UI State
Gemini AI API Outage / Timeout	<code>catch</code> block in <code>runInvestigationPipeline()</code>	Activates rule-based deterministic triage engine; sets <code>isDegradedMode: true</code> .	Displays amber badge: "DEGRADED INVESTIGATION (FAIL-SAFE)".
Overpass API Mirror Failure	HTTP error or 8.0s <code>AbortController</code> timeout	Iterates through 4 mirror endpoints; falls back to empty evidence array if all fail.	Step marked failed; displays: "Overpass dataset query limited; safe degraded fallback."
Nominatim Geocoding Failure	Network error or empty results	Stores raw text; sets <code>lat: null</code> , <code>lng: null</code> , and <code>locationQuality: "unresolved"</code> .	MiniMap displays prompt to manually enter location or use GPS.
Cloud Firestore Offline / Missing	<code>!hasFirebaseConfig</code>		<code>!db</code> check or read exception
Unauthorized Endpoint Access	<code>isAuthorized()</code> returns <code>false</code>	Returns HTTP 401 Unauthorized or HTTP 403 Forbidden.	UI renders <code>AuthorityGate</code> passcode modal.

40. Observability, Telemetry & Logging Architecture

40.1 Logging Conventions VERIFIED IN CODE

- Server routes log errors with detailed error messages (`console.error("Investigation stream execution error:", error)`).
- Fallback events are logged cleanly as warnings (`console.warn("Firestore snapshot access limited... engaging REST sync fallback")`).
- Sensitive secrets (`PASSCODE` , `GEMINI_API_KEY`) are never output to server logs.

41. Performance Optimization Architecture

41.1 Performance Strategies VERIFIED IN CODE

- 5-Minute Spatial Caching:** `geospatialCache` in `src/lib/geospatial.ts` prevents redundant Overpass queries within ~11m precision.
- Unified Capability Querying:** Merges multiple capabilities into a single Overpass QL union request.
- Singleton Firestore Subscription:** Reference-counted listener prevents duplicate WebChannel network streams.
- SSE Streaming:** Emits partial investigation steps immediately, reducing perceived user latency.
- Shared Basemap Promise:** `useMapConfig` executes a single fetch promise shared across all map instances.

42. Automated Testing Architecture & Test Scripts

42.1 Test Execution VERIFIED IN CODE

The repository includes an end-to-end automated test script: `scripts/test_phase2.mjs`.

To execute:

```
npm run dev &
node scripts/test_phase2.mjs
```

42.2 Test Suite Coverage Matrix VERIFIED IN CODE

- `TEST_01` : Electrical Fire & 6 Trapped People → Validates `fire_suppression` , `power_grid_isolation` , `trauma_care` , negative check on water rescue, confidence ≥ 75 .
- `TEST_02` : Municipal Water Main Rupture → Validates `water_grid_isolation` , `public_works_clearing` , negative check on water rescue & trauma care.
- `TEST_03` : River Flood Inundation → Validates `swift_water_rescue` , `fire_suppression` , `evacuation_support` .
- `TEST_04` : Medical Cardiac Arrest → Validates `trauma_care` , `ems_transport` , negative check on heavy USAR.
- `TEST_05` : Road Blockage by Fallen Tree → Validates `public_works_clearing` , `traffic_perimeter` , negative check on trauma care.
- `TEST_06` : Power Grid Outage (Standard) → Negative check on critical facility backup.
- `TEST_07` : Power Grid Outage (Hospital Context) → Positive check on `critical_facility_backup` .
- `TEST_08` : Industrial Gas Main Rupture → Validates `hazmat_containment` , `gas_grid_isolation` , `evacuation_support` .

- **TEST 09** : Structural Slab Collapse → Validates `heavy_extrication_usar` , `trauma_care` .
- **TEST 10** : Contact Provenance Verification → Validates `OSM:phone` provenance and `hasDirectPhone` .
- **TEST 11** : Protected API Security → Validates 401 on unauthenticated access and 200 after passcode authorization.
- **TEST 12** : Incident Lifecycle & Persistence → Validates POST creation and authorized PATCH team assignment.

43. Functional Acceptance Test Matrix

ID	Scenario	Input Conditions	Expected Investigation	Expected Evidence / Output	Expected UI State
FAT-01	Commercial Fire	Fire narrative, 4 trapped workers, GPS coords	Evaluates <code>fire_suppression</code> , <code>trauma_care</code> , <code>power_grid_isolation</code> .	Mapped fire stations & hospitals within 8–10 km.	Red zone badge, ≥ 90 confidence, ranked fire stations with direct calling.
FAT-02	Water Pipe Rupture	16-inch main burst, 0 injured	Evaluates <code>water_grid_isolation</code> , <code>public_works_clearing</code> .	Water works depot & valve infrastructure.	Amber zone, no water rescue boats shown.
FAT-03	River Flood	4ft water, families on roofs	Evaluates <code>swift_water_rescue</code> , <code>fire_suppression</code> , <code>evacuation_support</code> .	Rescue stations & fire stations.	Red zone, high urgency, swift water rescue recommendation.
FAT-04	Cardiac Arrest	Collapsed passenger, CPR in progress	Evaluates <code>trauma_care</code> , <code>ems_transport</code> .	Hospitals and ambulance depots.	Immediate urgency, EMS transport recommendation.
FAT-05	Unauthorized Dashboard Access	Direct visit to <code>/dashboard</code> without session cookie	None	HTTP 401 / Intercepted by <code>AuthorityGate</code> .	Fullscreen passcode modal displayed.
FAT-06	Dispatcher Team Assignment	Authorized dispatcher selects team on console	PATCH <code>/api/incidents/:id</code>	Updates <code>responseTeam</code> and appends to <code>assignmentHistory</code> .	Incident card reflects assigned status and team pill.

44. Deployment Architecture & Production Readiness

44.1 Deployment Topology VERIFIED IN CONFIG

- **Platform:** Vercel / Node.js Serverless Container.
- **Node.js Target:** Node.js 20.x+.
- **Build Command:** `npm run build` (`next build`).
- **Production Start:** `npm run start` (`next start`).
- **Dynamic Routes:** All API routes are serverless dynamic functions (`export const dynamic = "force-dynamic"` where required).

45. Developer Onboarding & Local Setup Guide

45.1 Quickstart Instructions

1. Clone Repository:

```
git clone <repo_url>
cd gdg
```

2. Install Dependencies:

```
npm install
```

3. Configure Environment:

```
cp .env.example .env.local
# Edit .env.local and add GEMINI_API_KEY, PASSCODE, etc.
```

4. Launch Development Server:

```
npm run dev
```

5. Access Web Application: Open `http://localhost:3000` in browser.

6. Run Automated Test Suite:

46. Data Storage & Persistence Contracts

46.1 Cloud Firestore Collection Structure VERIFIED IN CODE

- **Collection Name:** incidents
- **Document ID:** Auto-generated Firestore ID (or inc-<timestamp> in fallback).
- **Document Fields:** Conforms exactly to the Incident interface in src/types/incident.ts.
- **Query Indices:** orderBy("updatedAt", "desc").

47. External Service Contracts & SLA Limits

47.1 External Services Matrix VERIFIED IN CODE

Service Provider	Role	Protocol / Endpoint	Timeout	Error Handling
Google Gemini AI	Structured Diagnostic Reasoning	HTTPS / @google/genai	Default SDK	Degraded fail-safe rule engine.
OpenStreetMap Overpass	Geospatial Capability Queries	POST https://*/api/interpreter	8.0 seconds	4-mirror failover + spatial cache.
OpenStreetMap Nominatim	Forward Geocode & Reverse GPS	GET https://nominatim.openstreetmap.org/*	Standard	Fallback to raw text coordinates.
CARTO Basemaps	Dark Matter Map Tiles	HTTPS PNG Tile URLs	Browser	Default fallback dark basemap.
Google Cloud Firestore	Realtime Document Sync	gRPC / WebChannel	SDK default	Circular in-memory fallback.

48. Privacy, PII & Data Handling Policies

48.1 Data Handling Specification VERIFIED IN CODE

- **Citizen Report Text:** Processed through Gemini AI solely for crisis triage classification.
- **Geographic Coordinates:** Sent to Nominatim and Overpass to discover civic infrastructure; no private home ownership databases are queried.
- **Contact Details:** Mapped strictly from public OpenStreetMap tags (phone , website).

49. AI Safety, Epistemic Grounding & Anti-Hallucination

49.1 Grounding Safeguards VERIFIED IN CODE

1. **Schema Enforcement:** Strict JSON schema prevents unstructured prose hallucination.
2. **Epistemic Distinctions:** Prompts enforce explicit separation between **Facts**, **Inferences**, and **Unknowns**.
3. **No Phantom Units:** Resource candidates come strictly from Overpass API query results.
4. **Deterministic Confidence:** Numeric confidence is calculated strictly by mathematical logic in confidenceCalculator.ts.
5. **Human Dispatch:** Final operational decisions require explicit human dispatcher confirmation.

50. System Limitations & Known Constraints

50.1 Explicit Technical Limitations VERIFIED IN CODE

1. **OpenStreetMap Data Completeness:** In rural or unmapped areas, emergency facilities may be missing from the public OSM dataset.
2. **No Real-Time Staffing Feeds:** The platform does not have live telemetry on active paramedic shift rosters or hospital bed availability.
3. **Haversine vs Driving Distance:** Distances represent straight-line physical proximity, not road driving times.
4. **Ephemeral Memory Fallback:** In-memory fallback mode resets on server process restart.

51. System Threat Model

Threat Description	Impact Level	Mitigation Implemented	Residual Risk
Unauthorized Access to Command Center	High	<code>AuthorityGate</code> + HMAC-SHA256 <code>httpOnly</code> cookie verification.	Risk of shared/weak passcode if not rotated.
Passcode Timing Attack	Moderate	<code>crypto.timingSafeEqual</code> in <code>src/lib/auth.ts</code> .	Negligible.
AI Prompt Injection via Narrative	Moderate	Strict structured JSON schema validation and zero-execution sandbox.	Potential influence on reasoning text; does not affect deterministic resource queries.
Overpass API Rate Limiting / DoS	Moderate	5-minute spatial cache + 4 redundant global mirror endpoints.	Temporary degraded mode if all global mirrors are saturated.

52. Architectural Decision Records (ADRs)

52.1 Key Architectural Decisions VERIFIED IN CODE

- **ADR-01: Deterministic Confidence Calculation vs LLM Self-Scoring**
 - *Decision:* Compute confidence using mathematical algorithms in `confidenceCalculator.ts`.
 - *Rationale:* LLMs suffer from severe calibration bias and hallucinations when self-scoring confidence.
- **ADR-02: Server-Sent Events (SSE) for Investigation Telemetry**
 - *Decision:* Use SSE via Web Streams API (`ReadableStream`) over WebSockets.
 - *Rationale:* Unidirectional server-to-client streaming fits the investigation workflow with zero WebSocket connection management overhead.
- **ADR-03: Separation of Relevance Ranking vs Usability Flags**
 - *Decision:* Rank resources by capability match and distance; treat contact availability as usability indicators.
 - *Rationale:* Prevents demoting the nearest fire station simply because its phone number was omitted in OpenStreetMap.

53. Technical Debt & Codebase Enhancement Opportunities

53.1 Verified Enhancement Opportunities VERIFIED IN CODE

1. **Dynamic Radius Expansion:** If 0 resources are returned within 8 km, automatically expand query to 25 km before marking evidence missing.
2. **Turn-by-Turn Routing Engine:** Integrate OSRM / OpenRouteService for driving-time calculation alongside Haversine distance.
3. **Multi-User Auth Profiles:** Upgrade single passcode model to role-based access control (RBAC) with individual responder accounts.

54. Future Extensibility Roadmap

54.1 Extension Points NOT IMPLEMENTED

- **Municipal CAD Integration:** Direct bidirectional API uplink with municipal Computer-Aided Dispatch (CAD) systems.
- **WhatsApp / SMS Incident Intake:** Webhook ingestion of civilian incident reports from messaging platforms.
- **Responder Field PWA:** Dedicated mobile progressive web app with offline caching and GPS field unit tracking.

55. Developer Maintenance Guide ("How to Change X")

55.1 How to Add a New Incident Category

1. Update `IncidentCategory` union in `src/types/incident.ts`.
2. Add category to `CATEGORY_LABELS` in `src/app/report/page.tsx`.
3. Add scoping rules in `planRequiredCapabilities()` in `src/lib/investigationEngine.ts`.
4. Update `triageReasoningSchema` enum in `src/lib/investigationEngine.ts`.
5. Add acceptance test scenario in `scripts/test_phase2.mjs`.

55.2 How to Add a New Resource Capability

1. Add capability key to `ResourceCapability` union in `src/types/investigation.ts`.
2. Add entry to `CANONICAL_CAPABILITY_REGISTRY` in `src/lib/resourceRanker.ts`.

3. Add Overpass query clauses to `buildCapabilityOverpassQuery()` in `src/lib/geospatial.ts`.
4. Update mapping logic in `normalizeOverpassElements()` in `src/lib/geospatial.ts`.

56. Requirements-to-Code Traceability Matrix

Requirement ID	Module / File	Primary Export / Symbol	Verification Test
FR-INT-001	<code>src/app/report/page.tsx</code>	<code>ReportPage</code>	Manual / E2E
FR-INT-002	<code>src/app/report/page.tsx</code>	<code>searchPlaces</code> , <code>handleDetectLocation</code>	Manual / E2E
FR-INV-001	<code>src/lib/investigationEngine.ts</code>	<code>runInvestigationPipeline</code>	<code>test_phase2.mjs</code> (Tests 1-9)
FR-INV-002	<code>src/lib/investigationEngine.ts</code>	<code>planRequiredCapabilities</code>	<code>test_phase2.mjs</code> (Tests 1-9)
FR-GEO-001	<code>src/lib/geospatial.ts</code>	<code>searchNearbyCapabilities</code>	<code>test_phase2.mjs</code> (Tests 1-10)
FR-RNK-001	<code>src/lib/resourceRanker.ts</code>	<code>rankOperationalResources</code>	<code>test_phase2.mjs</code> (Tests 1, 10)
FR-CNF-001	<code>src/lib/confidenceCalculator.ts</code>	<code>calculateDeterministicConfidence</code>	<code>test_phase2.mjs</code> (Test 1)
FR-AUT-001	<code>src/lib/auth.ts</code>	<code>verifyPasscode</code> , <code>isAuthorized</code>	<code>test_phase2.mjs</code> (Test 11)
FR-CMD-001	<code>src/app/dashboard/page.tsx</code>	<code>DashboardPage</code>	Manual / E2E
FR-LIF-001	<code>src/app/api/incidents/[id]/route.ts</code>	<code>PATCH</code>	<code>test_phase2.mjs</code> (Test 12)

57. Reverse File-to-Feature Lookup Matrix

Feature Area	Primary Source Files Involved
Evidence-Based AI Triage	<code>src/lib/investigationEngine.ts</code> , <code>src/lib/ai.ts</code> , <code>src/app/api/triage/route.ts</code> , <code>src/app/api/triage/stream/route.ts</code>
Deterministic Confidence	<code>src/lib/confidenceCalculator.ts</code> , <code>src/types/investigation.ts</code>
Geospatial & Overpass Discovery	<code>src/lib/geospatial.ts</code> , <code>src/lib/resourceRanker.ts</code> , <code>src/app/api/resources/nearby/route.ts</code>
Authentication & Gate	<code>src/lib/auth.ts</code> , <code>src/contexts/AuthorityContext.tsx</code> , <code>src/components/shared/AuthorityGate.tsx</code> , <code>src/app/api/auth/authority/route.ts</code>
Incident Command Console	<code>src/components/shared/IncidentDetailPanel.tsx</code> , <code>src/components/shared/IncidentCard.tsx</code> , <code>src/app/dashboard/page.tsx</code>
Interactive Tactical Mapping	<code>src/components/shared/LiveMap.tsx</code> , <code>src/components/shared/MiniMap.tsx</code> , <code>src/components/shared/MapWrapper.tsx</code> , <code>src/hooks/useMapConfig.ts</code> , <code>src/app/api/map-config/route.ts</code>
Realtime State & Persistence	<code>src/lib/firebase.ts</code> , <code>src/hooks/useIncidents.ts</code> , <code>src/lib/memoryStore.ts</code> , <code>src/app/api/incidents/route.ts</code> , <code>src/app/api/incidents/[id]/route.ts</code>

58. Formal Requirements Classification & Unique IDs

58.1 Functional Requirements (FR)

- FR-001 (Citizen Report Intake):** The system shall accept incident reports with title, narrative description, category tag, and casualty estimates.
- FR-002 (Geocoding & Location Quality):** The system shall evaluate location quality (`exact_gps`, `resolved_address`, `approximate_city`, `unresolved`) using Nominatim and HTML5 Geolocation.
- FR-003 (Capability Planning):** The system shall deterministically map incident profiles to required capabilities using positive and negative scoping rules.
- FR-004 (Overpass Geospatial Search):** The system shall query Overpass API mirrors for verified emergency facilities within tactical radii (4-15 km).
- FR-005 (Contact Provenance Extraction):** The system shall extract OSM contact tags (`phone`, `website`, `addr:*`) with field-level provenance (`OSM:phone`).
- FR-006 (Deterministic Resource Ranking):** The system shall rank discovered resources based on capability match (50), proximity (40), and context (10).
- FR-007 (Gemini Evidence Reasoning):** The system shall synthesize verified OSM evidence and narrative facts using Gemini 2.5 Flash with strict JSON schema enforcement.

- **FR-008 (Deterministic Confidence Calculation):** The system shall compute multi-dimensional confidence (C, S, E, L) using server-side mathematical formulas clamped to [15, 99].
- **FR-009 (SSE Live Investigation Streaming):** The system shall stream real-time investigation events via `POST /api/triage/stream`.
- **FR-010 (Human Dispatch Authority):** The system shall require explicit dispatcher approval to assign teams, override priorities, and resolve incidents.

58.2 Non-Functional Requirements (NFR)

- **NFR-001 (Confidence Compute Latency):** Mathematical confidence scoring shall execute in < 5 ms.
- **NFR-002 (Overpass Query Timeout):** Geospatial requests shall be bounded by an 8.0-second abort timeout per mirror.
- **NFR-003 (Spatial Cache TTL):** Spatial query results shall be cached in memory for 300 seconds (5 minutes).
- **NFR-004 (Cryptographic Session Expiration):** HMAC-signed authority session cookies shall expire after 7 days (604,800 seconds).

58.3 Security Requirements (SEC)

- **SEC-001 (Command Center Authorization):** All access to `/dashboard`, `/api/resources/nearby`, and `PATCH /api/incidents/:id` shall require valid `authority_session` cookies.
- **SEC-002 (Timing Attack Prevention):** Passcode verification shall use `crypto.timingSafeEqual`.
- **SEC-003 (Cookie Hardening):** Session cookies shall be flagged `httpOnly: true`, `sameSite: "lax"`, and `secure: true` in production.

59. Non-Functional Requirements (NFR) Specification

59.1 Measurable Benchmarks VERIFIED IN CODE

- **Overpass Query Timeout:** Maximum 8000 ms per mirror endpoint.
- **Spatial Cache Expiration:** 300,000 ms (5 minutes).
- **Session Max Age:** 604,800 seconds (7 days).
- **REST Fallback Polling Interval:** 4000 ms (when Firestore listener is unavailable).
- **Firestore Debounce Teardown:** 1000 ms grace period on component unmount.

60. Project Glossary & Domain Terminology

- **AI-Assisted Triage:** Decision-support process where AI extracts facts, queries infrastructure, and recommends priorities, leaving final authority to human dispatchers.
- **Canonical Capability Registry:** Formal catalog of 13 standardized emergency response and utility isolation capabilities.
- **Contextual Infrastructure:** Physical assets (substations, pipelines, water works) evaluated for environmental risk assessment but never deployed as emergency response units.
- **Deterministic Confidence:** Transparent point-based mathematical scoring of incident quality, severity clarity, evidence corroboration, and location precision.
- **Emergency Response Resource:** Deployable municipal rescue units (fire engines, ambulances, police patrols, USAR squads).
- **Epistemic State:** Explicit categorization of operational information into verified Facts, tactical Inferences, and acknowledged Unknowns.
- **Overpass API:** Read-only API serving custom selected subsets of OpenStreetMap geospatial data.
- **Proximity Score:** Exponential decay score $P(d) = 100 \times \exp(-0.15 \times d)$ mapping physical distance to response relevance.
- **Severity Zone:** Tactical tri-color classification: Red (Score 80–100, Critical Life Safety), Amber (Score 45–79, Major Hazard), Green (Score 0–44, Stable/Minor).
- **Zero-Hallucination Contact Extraction:** Protocol ensuring all telephone numbers, websites, and addresses originate strictly from verified OpenStreetMap tags with provenance tracking (`OSM:phone`).

End of Software Requirements Specification (SRS-NEEDNOW-2026-V1.0)